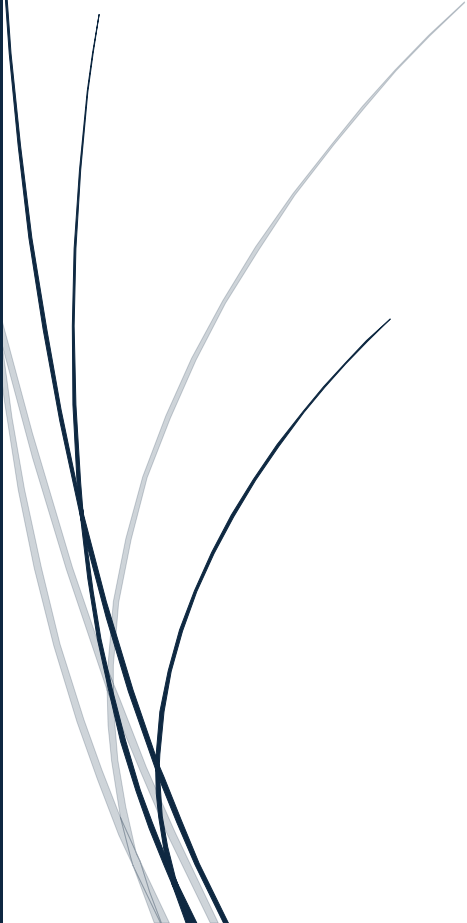


A dark blue vertical bar on the left side of the page. A horizontal arrow, also in dark blue, points to the right from the bar, containing the date.

05/05/2026

Demographic Breakdown Analysis

Dataset Used: Adult Psychiatric Morbidity Survey (APMS)
2014

Several thin, curved lines in dark blue and light grey originate from the bottom left corner and sweep upwards and to the right.

Prepared By: Mukesh Raj
Data Analytics Intern

1. Introduction

Demographic breakdown analysis was conducted to examine how the prevalence of mental health disorders varies across different population segments. The focus of this analysis was on **age groups** and **gender categories**, using grouped visualizations to identify patterns and trends within the dataset.

2. Methodology

To perform the analysis, the dataset was first transformed into a long format using data reshaping techniques. This allowed for easier comparison across multiple variables such as age group, gender, and disorder categories.

The analysis included:

- I. Grouping data by age bands (**16–24 to 75+**)
- II. Comparing across gender groups (**Men, Women, All Adults**)
- III. Evaluating distribution across different **mental health categories**

For visualization, **grouped bar charts** were used to clearly represent differences between demographic groups.

3. Age-wise Analysis

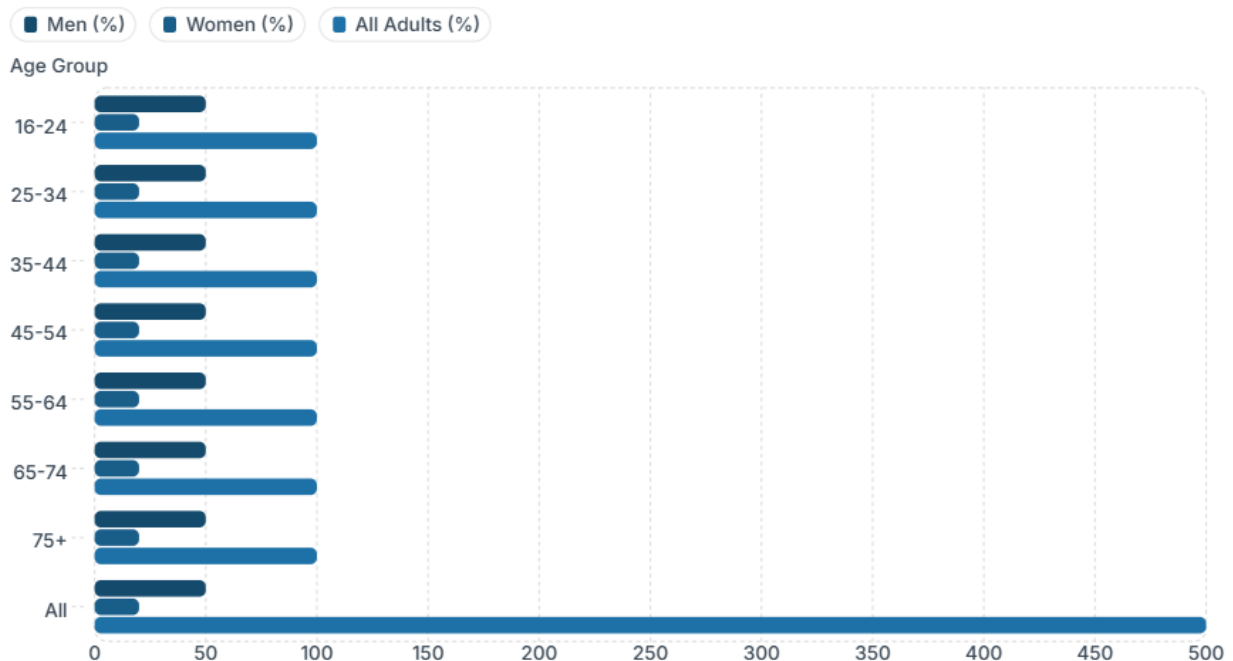
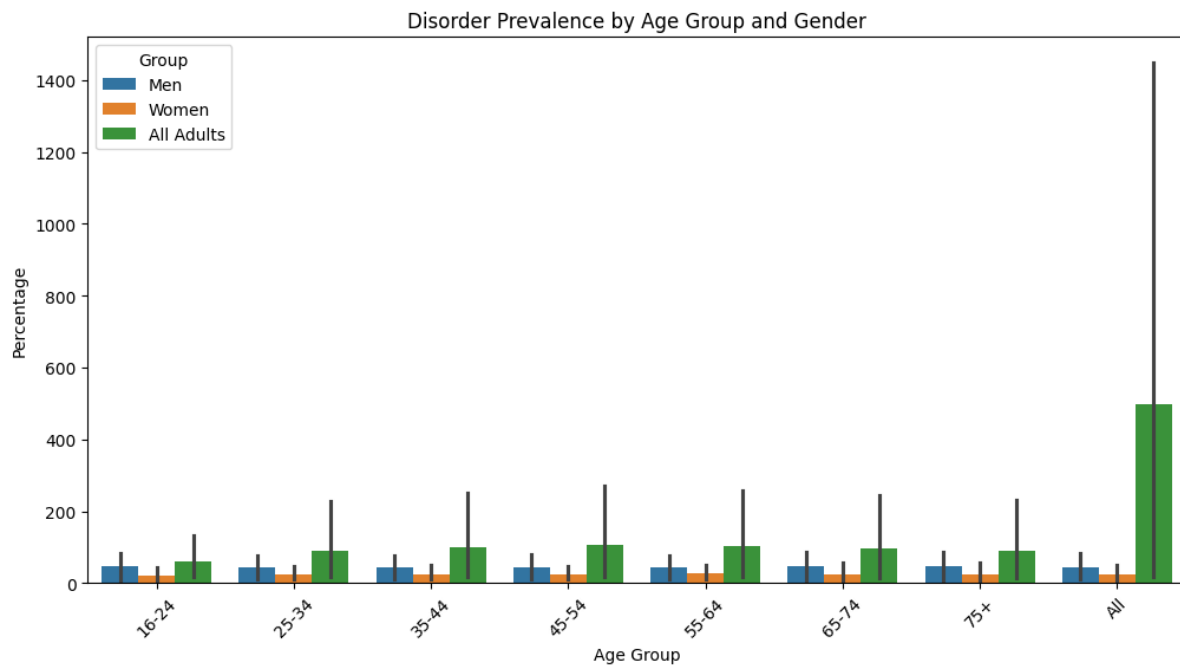
The analysis across age groups reveals that mental health disorder prevalence is not uniform across all age bands. Certain age groups show relatively higher percentages, indicating that mental health conditions may be more prominent within specific stages of life.

Younger and middle-aged groups tend to show more variability in values, while older age groups may exhibit comparatively stable or lower prevalence in certain categories.

```
# Age-wise Analysis
plt.figure(figsize=(12,6))
sns.barplot(data=df_melted, x="Age Group", y="Percentage", hue="Group")

plt.title("Disorder Prevalence by Age Group and Gender")
plt.xlabel("Age Group")
plt.ylabel("Percentage")
plt.xticks(rotation=45)

plt.show()
```



4. Gender-wise Analysis

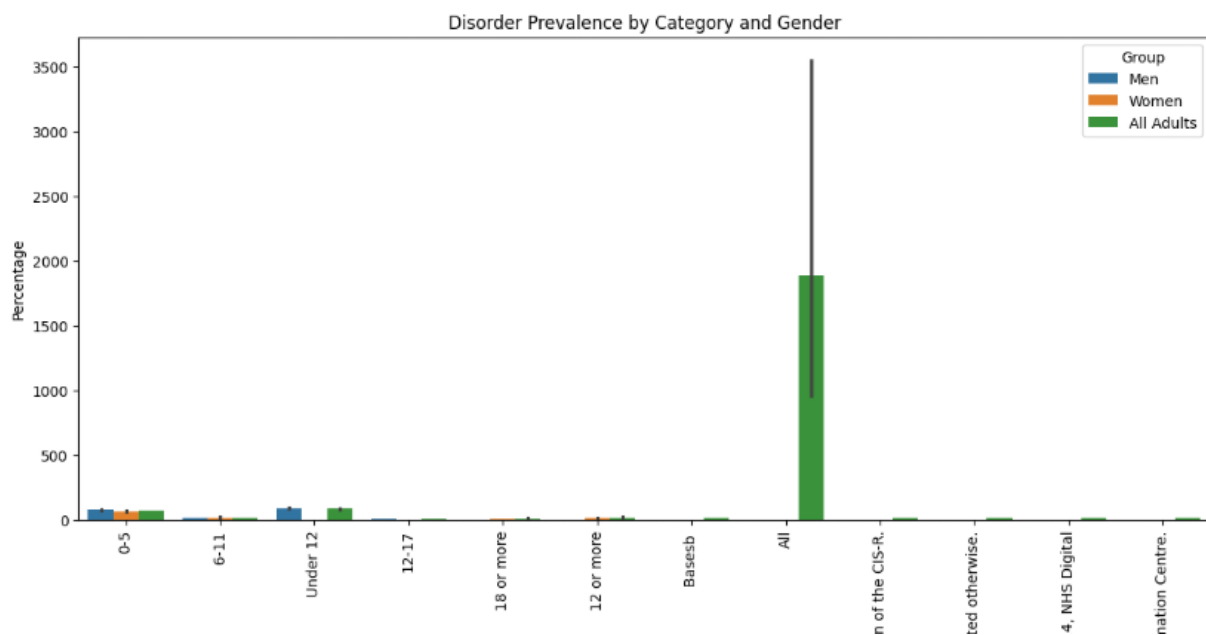
Gender-based comparison highlights noticeable differences between **Men** and **Women** across multiple disorder categories. In several cases, one gender demonstrates higher prevalence rates than the other, suggesting potential social, biological, or environmental influences on mental health conditions.

The inclusion of the “**All Adults**” category provides an aggregated view, helping to validate overall trends observed in individual gender groups.

```
plt.figure(figsize=(14,6))
sns.barplot(data=df_melted, x="Category", y="Percentage", hue="Group")

plt.title("Disorder Prevalence by Category and Gender")
plt.xlabel("Category")
plt.ylabel("Percentage")
plt.xticks(rotation=90)

plt.show()
```



Category	Men (%)	Women (%)	All Adults (%)
0-5	~100	~100	~100
6-11	~10	~10	~10
Under 12	~150	~10	~150
12-17	~10	~10	~10
18 or more	~10	~10	~10
Baseline	~10	~10	~10
All	~10	~10	~1900

5. Category-wise Distribution

When analyzing disorder categories, it is evident that different mental health conditions exhibit varying prevalence across demographic groups. Some categories show consistently higher values across all groups, while others vary significantly depending on age and gender.

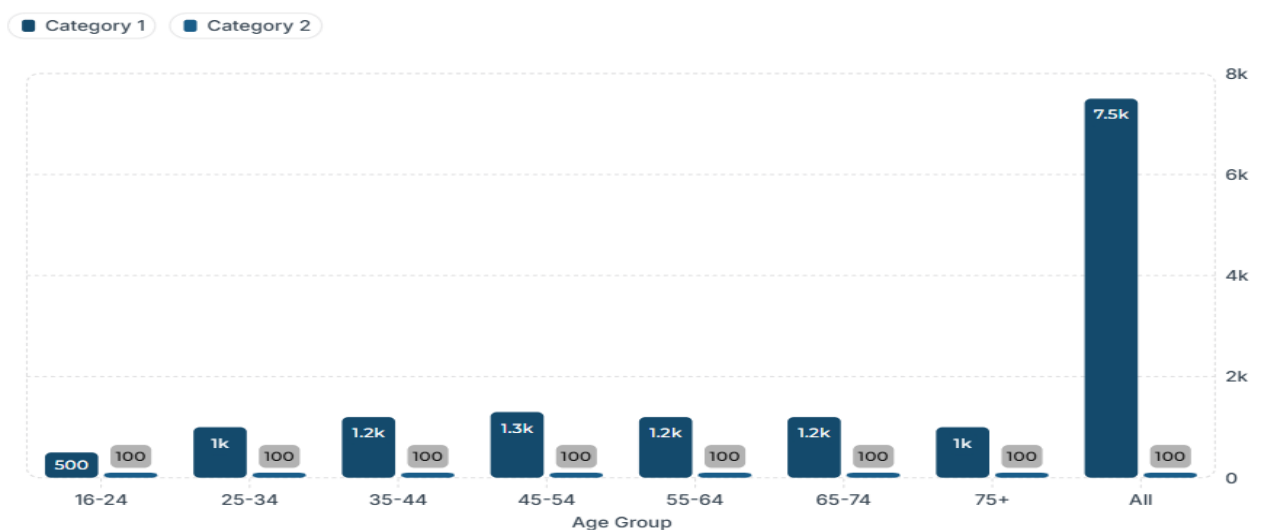
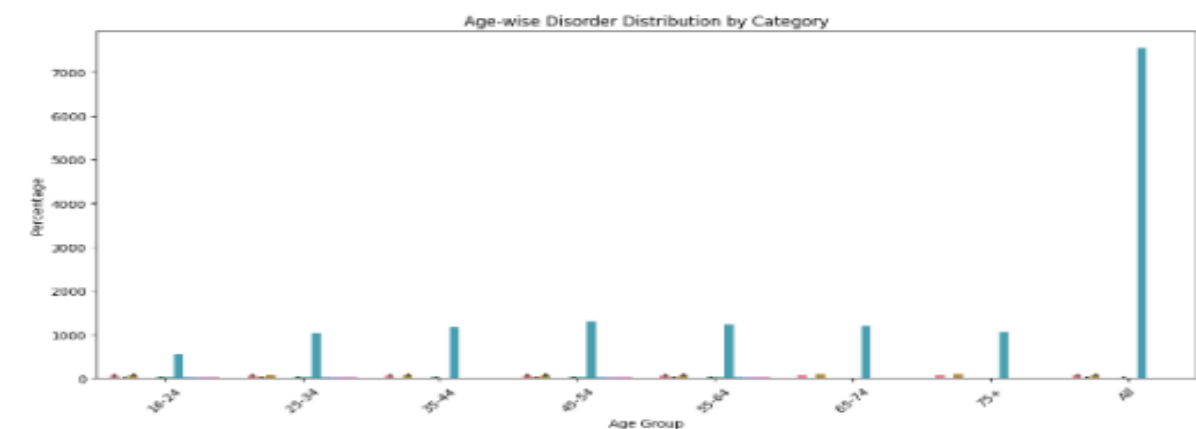
This variation emphasizes the importance of demographic segmentation in understanding mental health patterns.

```
# Age vs Category
plt.figure(figsize=(14,6))
sns.barplot(data=df_melted, x="Age Group", y="Percentage", hue="Category")

plt.title("Age-wise Disorder Distribution by Category")
plt.xlabel("Age Group")
plt.ylabel("Percentage")
plt.xticks(rotation=45)

plt.legend(bbox_to_anchor=(1.05,1), loc='upper left')

plt.show()
```



6. Ethnicity Analysis

Ethnicity-based analysis could not be performed due to the **absence of ethnicity-related variables** in the dataset. Inclusion of such data would enable a more comprehensive demographic assessment.

7. Conclusion

The demographic breakdown analysis highlights that mental health disorder prevalence varies significantly across **age groups and gender categories**. The findings demonstrate the importance of demographic factors in understanding mental health trends.

The use of grouped bar charts effectively illustrates these variations, supporting better interpretation and decision-making. However, the absence of ethnicity data limits the scope of the analysis, suggesting an area for future enhancement.